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JOHNS HOPKINS APL PARADIGM IN SMALLSAT MANAGEMENT

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ABSTRACT

The Johns Hopkins University Applied Physics Laboratory (JHU/APL) has designed, built, flown, and operated 57 satellites over the past 40 years, utilizing the methodology currently captured by the phrase "better, faster, smaller, cheaper" for all but one of these missions. In so doing, a management culture evolved that resulted in successful meeting of technical requirements within short schedules, often below originally estimated costs. The basic guidelines used in this approach are presented and illustrated with examples from recent or ongoing programs. The practical implementation of these guidelines varies from one program to another. A case in point is the recent trend to minimize mission operations costs. A good example of innovative architecture is the one employed on TIMED. It combines a Global Positioning System (GPS)–linked transponder, to enable time, velocity, and position for data tagging and autonomous orbit determination and propagation, with a ground system that enables separate operation of instruments from any of the spacecraft subsystems. © 2000 Elsevier Science Ltd. All rights reserved

INTRODUCTION

Spacecraft development began at The Johns Hopkins University Applied Physics Laboratory (JHU/APL) some 40 years ago, following the discovery by two JHU/APL scientists [1] of the principle of the Doppler method for determining the orbital elements of an Earth-orbiting spacecraft. Prior to this seminal discovery, the prevailing mode of orbit determination used by nearly everyone was angle–angle tracking from multiple stations, which resulted in poor determination of orbital parameters and substantial errors. The Doppler method concept was immediately inverted to that of a satellite broadcasting its orbital elements with the use of Doppler signals to determine the location of an observer on the surface of the Earth (memo of F. T. McClure, 18 March, 1958), with obvious applications for navigation of military and civilian vehicles alike.

Transit Navigation System

The first satellite launch to demonstrate this principle occurred in 1959 and, despite the failure to achieve orbit, the ascent data were used to demonstrate that such navigation was possible. The go-ahead was obtained with Navy funding to proceed with implementation of an operational system. Within the next 4 years, a small constellation of spacecraft, consisting of four satellites orbiting at an altitude of about 1,000 km and inclination

of 90°, was already providing navigation signals for ocean and land-based users with a precision of about 1 km and repeat time of approximately 1.5 hours [2].

The remarkably short time between initial concept and operational availability was achieved by a dedicated group of scientists and engineers who were able to design, build, launch, and operate some 20 spacecraft (seven of which did not reach orbit) in a period of approximately 4 years. Such an accomplishment is thought to be impossible today, in spite of the intervening 40 years of technological progress. Once the Transit navigation system was declared operational, the technology developed in its implementation was transferred to industry (RCA Space Systems) and a series of operational spacecraft were built and warehoused for replacing those in the constellation that may have failed.

The Transit system, officially retired at the end of 1996, compiled one of the most remarkable records of reliability (99.9% operational with 14-year mean-time-to failure in orbit) and of longevity (Oscar 13 holds the record for the longest operational spacecraft at 21.7 years) of any operational system designed and built in the 1960s.

Implementation of the Transit system resulted in a large number of technical innovations that have become standard designs ever since in the satellite industry worldwide. These include such techniques as the yo-yo despin mechanism (1959), the two-frequency method for correcting ionospheric errors (1960), the first attitude-controlled spacecraft using permanent magnets (1960), hysteresis damping of satellite libration (1960), gravity-gradient stabilization (1963), the first Radioisotopec Thermoelectric Generator (RTG) in space (1961), and the first drag-free satellite (1972), to name just a few [3].

The engineering innovations were accompanied by an equivalent thrust in the evaluation and modeling of the space environment, including the first determination of Earth's geoid [4]. Scientific payloads were designed and flown into low-Earth orbit to assess the impact of the Van Allen radiation particles on spacecraft electronics, to measure the intensity of solar flare-emitted protons over the Earth's polar caps, and to investigate disturbances associated with spacecraft attitude in northern and southern latitudes [5]. The last effect was eventually traced to the discovery of field-aligned currents (FAC) flowing along magnetic field lines in the auroral zone. The FACs have been shown to play a fundamental role in the electrical circuit of the ionosphere–magnetosphere system and are considered part of the fundamental properties of plasmas, not only in Earth and planetary magnetospheres, but also in the solar corona and the heliosphere at large.

The NASA Missions Era

The impressive performance of APL's scientists and engineers on the Transit system attracted the attention of other organizations, including NASA, who needed help in the implementation of one-of-a-kind, technically challenging missions in astronomy and space physics. APL undertook the task of designing and building a series of Small Astronomy Satellites (SAS) in the early 1970s, the first of which was the Uhuru x-ray satellite, which performed the first survey of x-ray sources in the sky, with R. Giaconi as principal investigator.

The SAS missions were followed by the first-ever radar altimeter spacecraft, launched in 1976 (Geos-C); the first magnetic field-mapping satellite (MAGSAT); and the three-spacecraft international Active Magnetospheric Particle Tracer Explorers (AMPTE) Program, a collaboration between the United States, Germany, and the United Kingdom that performed the first release of an artificial comet in space [6].

The decade of the late 1970s to mid-1980s had witnessed the evolution of a much different approach from that of the 1960s in the conduct of space science missions. Programs were now being planned that took 10 years to implement, and costs had begun to escalate from tens of millions into hundreds of millions of dollars. Simultaneously, the frequency of missions had decreased drastically, and timely implementation of state-of-the-art scientific missions had become nearly impossible. The space science and technology community had begun a search for a more efficient and less costly way of implementing missions. This search was intensified in the late 1980s and early 1990s and culminated in the establishment of the Discovery Program in the United States [7] and the introduction of the "better-faster-cheaper-smaller" rubric by NASA administrator Dan Goldin.

Inevitably, in this search for efficiency, the management methodology practiced by APL over the preceding 30 years attracted considerable attention, since it had become the embodiment of Mr. Goldin's rubric, without actually having been enunciated as such. The result was the selection of APL to implement the first Discovery mission, the Near Earth Asteroid Rendezvous (NEAR) to orbit the asteroid Eros.

The NEAR mission was designed to achieve the basic objectives set by the Discovery Program: (a) a goal of $150 million (fiscal year 1992 dollars); (b) implementation in less than 36 months; and (c) launch by a medium-sized launch vehicle (Delta equivalent or smaller). All of the above were easily attained by NEAR, which was completed in record time of 27 months for a planetary mission, cost $104 million, and was launched by a Delta rocket. The NEAR rendezvous with Eros is now scheduled to take place in February 2000.

The paradigm for management of such programs can be summarized by the following seven management guidelines [8]:

1. Schedule from start to launch must be ≤36 months;
2. Establish a small, experienced technical team;
3. Design spacecraft and instruments to cost;
4. Use the lead engineer method for all subsystems;
5. Design in reliability and redundancy at the outset;
6. Integrate the product assurance engineer into the program;
7. A single agency manager is assigned to interface with the development teams.

We will discuss each of these guidelines briefly and highlight its significance in terms of cost and performance.

The new paradigm is here to stay. It is practiced widely by a large number of national space agencies, agency development centers, universities, and industry. Nevertheless, the pressure on reducing overall mission costs continues, especially for space and Earth science missions. The area that is now receiving most of the

attention is that of mission operations (MO). In this context, we describe the MO plan for the NASA Thermosphere-Ionosphere-Mesosphere Energetics and Dynamics (TIMED) mission. We demonstrate how it is possible to use a Global Positioning System (GPS)–linked transponder with the data to allow more-or-less autonomous operations that result in drastically reduced mission operations costs.

MANAGEMENT GUIDELINES

Schedule from Start to Launch Should Be Less Than 36 Months

The 36-month period is based on the typical mission for JHU/APL. The time period may be different based on the type of mission typical for your organization. The primary issue is to fix the development phase such that the proper amount of time is allowed for study, design, fabrication, and test. Too much time in the study and design phase feeds the concept that "better is the enemy of good enough" and raises the cost of the eventual product in both time and money. Additional time in the latter phases of development means a "standing army" is being extended, which is very expensive and also tends to defocus the team.

Establish a Small, Experienced Technical Team

The size of a development team is based on schedule and technical complexity. The important element is that the size of the team allows the requirements to be accomplished based on previous successful mission experience with respect to cost and schedule. One important factor is to maintain a realistic challenge in design. That is, the manager should challenge the designers with the schedule, but they must recognize that it is achievable or frustration will replace challenge. They must also realize that only through an innovative approach is it achievable, once again maintaining focus throughout the team.

The experience aspect is that decisions must be made within the team whenever possible; this will do a lot for bettering team dynamics. In looking at the experience level of the team, it is also important that management be part of the grooming process for future lead engineers and system engineers. There needs to be an appreciable sprinkling of early career staff behind each lead engineer. This is primarily the responsibility of the skill center supervisor, but the project manager must share in that responsibility for the improvement of the organization.

Design Spacecraft and Instruments to Cost

At the onset of the development phase, there needs to be a complete understanding of the mission requirements as viewed by the customer, science team, and spacecraft team. Equally important are the derived requirements based on the mission requirements: such items as spacecraft pointing, data rates, processor architecture, etc. Again there needs to be discussion and agreement that all mission requirements are achieved but not exceeded using the list of derived requirements.

Once the mission and derived requirements are in place, the cost of the project can be established. In cost-capped projects, the requirements may need to be iterated, as cost is the fixed parameter. It is even more important that there be agreement between the customer and the development team over the resulting mission requirements. The single highest risk

to managing a cost-capped program is "requirements creep."

From a management viewpoint there needs to be focus on all three of the aforementioned groups (the customer, the science team, and the development team). As the development phase progresses, the customer will obviously desire to obtain as much "mission" as can be acquired for the cost. The science team will assuredly desire the best possible instruments that money can't buy. Finally, all of the subsystem engineers developing the spacecraft are interested in challenging themselves—they're engineers.

Continued focus on the cost-capped design must be a part of all presentations, meetings, and discussions. It is very important to prevent any creep in the original agreements. Once the various elements of the development team recognize that the requirements are fixed, and that will take time, the job becomes easier. Conversely, if management shows signs of listening and perhaps even incorporating better ideas, rest assured that a continued diet of "better ideas" will be on the plate and just assuredly cause cost problems.

Table 1 shows the cost performance of recent JHU/APL spacecraft buses developed using this approach.

Use the Lead Engineer Method for Each Subsystem

The JHU/APL Space Department is organized in matrix format. The project office consists of a Project Manager,

Table 1. Spacecraft Bus Recent History.

Program	Sponsor	Weight of Bus, kg Dry Mass	Launch Date	Program Duration, Months	Constant FY 97* $M	Cost* Growth,** %
MAGSAT	NASA/GSFC	158	1979	24	26.3	8
AMPTE	NASA/GSFC	156	1984	36	27.0	0
GEOSAT-A	NAVY	543	1985	37	33.2	7
POLAR BEAR	USAF	94	1986	24	18.0	–5
Delta 180	SDIO	323	1986	12	20.6	–3
Delta 181	SDIO	1081	1988	17	114.7	–1
Delta 183	SDIO	253	1989	13	21.8	8
NEAR	NASA/HQ	429	1996	26	71.9	–7
MSX***	BMDO	1449	1996	40	112.0	13
ACE	NASA/GSFC	590	1997	47	48.8	–10

*Cost at project completion (launch +30 days) in constant fiscal year (FY) dollars

**Percent cost growth from initial cost estimate at start of Phase C/D

***Cost adjusted for 3.5 years of program schedule slip—funding shortfall, government-furnished instrument delays, programmatic delays

System Engineer, and Secretary. The staff is in skill centers such as telecommunications, mechanical, software, etc. At the start of a project, the Project Manager and System Engineer meet with the supervisor of each skill center and the lead engineer assigned by the skill center supervisor. Essentially, a contract is agreed upon between each skill center and the project. This contract includes technical requirements, schedule, and cost. The lead engineer is then responsible for fulfilling the contract for all three of the above issues (technical, schedule, and cost). It is incumbent upon both the skill center supervisor and the Project Manager to periodically discuss the performance of the lead engineers.

This approach provides a sense of ownership and therefore the desired responsibility in the lead engineer. The lead engineer also usually stays with the project through concept, development, fabrication, test, and launch. This approach also eliminates a significant amount of documentation, which is usually generated only because there are several handoffs when this approach is not utilized.

Design in Reliability and Redundancy at the Outset

Discussions are necessary in the early phase of a mission design between the customer, the science team, and the implementing agency (contractor) to establish the reliability requirements with respect to the spacecraft, instruments, and overall mission. These discussions allow the necessary redundancy to be "designed in" and not added on at a later and more expensive date. Often, functional redundancy can be attained at a lower cost than "block" redundancy (i.e., the duplication and necessary cross-strapping of a module).

Two examples of functional redundancy were utilized on the NEAR Program. One of the imaging instruments, the MultiSpectral Imager, was planned as a backup for the single-string star camera onboard the spacecraft. There would be a slight but tolerable degradation in pointing, but with early discussions, such tradeoffs can be managed. A second example on NEAR is the functional redundancy between the reaction wheels and the attitude control propulsion system. The inherent redundancy was considered in the design and cost of both subsystems.

Early recognition that block redundancy is the proper way to increase reliability allows proper harness design if cross-strapping is required.

Integrate the Product Assurance Engineer Into the Program

Many organizations structure the Reliability and Quality Assurance role outside the program so that it reports through the line organization to upper management. JHU/APL takes a different approach and integrates this role into the development team with the title "Product Assurance" (PA) engineer. With this approach the "cop on the beat" atmosphere is lessened; this engineer has a product to deliver as do all other subsystem engineers. Communication is stronger and discussion of the problems that do occur is more open. In the event that a Project Manager is not taking the appropriate action with respect to a particular issue, the PA engineer has a secondary path through the line organization to upper management. It is a rare

occasion when this secondary path is utilized.

A Single Agency Manager Is Assigned to Interface with the Development Team

This guideline is different from the rest in that it affects the organization of the customer. The rationale for this guideline will be illustrated by a hypothetical but possible example.
If JHU/APL is embarking upon a project with NASA as a customer, but perhaps involving more than one of the field centers, each center may have slightly varying requirements, such as in Product Assurance. What may appear to be slight variation within NASA could have potential cost and schedule impact upon the implementing team. It is essential to establish a lead center as the interface. If this does not occur, an agreed-upon set of rules for all issues must be established at the outset. After development is underway is not the time to discuss rules, mainly because the interfaces are then between engineers or scientists who typically do not set policy and may make agreements that cause schedule or cost impact.

As originally stated, this guideline is the one over which one may have least control, but it should be brought to the customer's attention early and should be a cause for concern.

MISSION OPERATIONS CONCEPT ON TIMED

The TIMED Program (Figure 1) is currently being developed as the first mission in NASA's Solar Terrestrial Probes

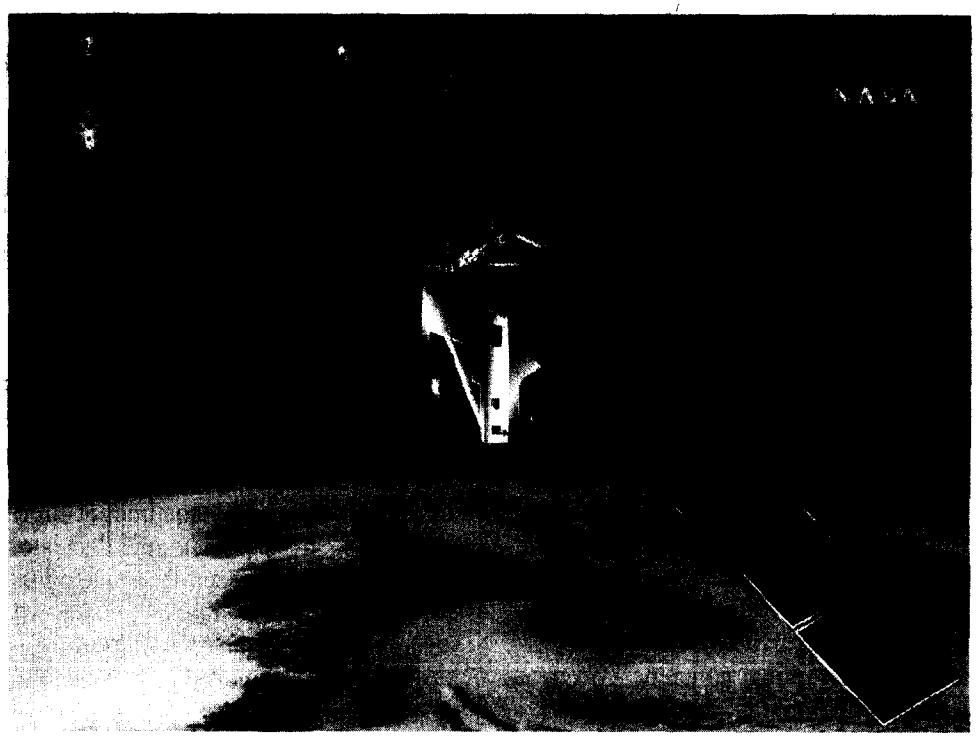


Figure 1. The TIMED spacecraft.

Program. TIMED is a low-cost mission aimed at providing a basic understanding of the least-explored and least-understood region of the Earth's environment—the segment of the Earth's atmosphere extending from 60 to 180 km in altitude. The TIMED suite of instruments is intended to determine the temperature, density, and wind structure in the mesosphere, lower thermosphere, and ionosphere (MLTI), including seasonal and latitudinal variations. TIMED is also intended to determine the relative importance of various radiative, chemical, electrodynamical, and dynamical sources and sinks of energy for the thermal structure of the MLTI.

The remainder of this paper discusses how the TIMED Program is using advanced technology to reduce Mission Operations and Data Analysis (MO&DA) costs [9].

Decoupled Operations and the "Greyhound Bus Operations Concept"

Over a series of meetings at JHU/APL in mid-1994, a simplified method of operating the TIMED spacecraft evolved that relied increasingly on automated processes, as well as distributed and decoupled operations, to reduce staffing requirements. Some of the early changes were obvious and simple to implement. The first such change was to reduce MO from 24-hour operations to single-shift, seven-day-per-week operations. This resulted in at least a three-fold reduction in the cost of MO. The second major change was less obvious. The "Greyhound Bus Operations Concept" originated as a metaphor for decoupled instrument operations. The argument was advanced that a Greyhound Bus is cheaper to operate than a spacecraft partly because nobody has to orchestrate all of the activities of the bus passengers. Passengers get on and off the bus, eat, sleep, and use the restroom without guidance or oversight from an omnipotent operations coordinator. The bus line is responsible only for maintaining the bus, keeping it on its designated route, handling en-route contingencies, and announcing stops and points of interest to the passengers, who use this information to guide their own decision-making processes.

Under this Concept of Operations (ConOps), the Mission Operations Center (MOC) and Operations Team, in the role of the bus line and driver, are responsible only for maintaining and operating the spacecraft bus. The Payload Operations Centers (POCs) and Instrument Teams, in the role of the passengers, are each responsible for planning and implementing their own operations. This decoupling of operations results in a considerable reduction in MO staffing. It eliminates the need for coordinating, negotiating, deconflicting, and vetting instrument command uploads. While this may seem simple and obvious, it is revolutionary in spacecraft operations.

It should be noted that this ConOps is not without cost or impact to the spacecraft. The Greyhound Bus ConOps can only be implemented if all spacecraft resources are sufficiently robust and abundant to eliminate the need to budget-limit resources such as power and data storage. The TIMED spacecraft is designed to provide sufficient resources for all instruments to operate at a 100% duty cycle. For TIMED, a conscious decision was made to trade the cost of increased spacecraft resources (power and data capacity) for decreased post-launch MO staffing. Note that in some programs, pointing or field of view is a limited resource, i.e.,

instruments have conflicting needs to point the spacecraft; in those situations, the Greyhound Bus paradigm cannot be applied as easily or successfully as envisioned for TIMED. The whole concept is illustrated in Figure 2.

Event-Based Commanding and the Global Positioning System

For purposes of this discussion, an "event" is defined as a detectable occurrence that can be used as a stimulus to initiate an activity or sequence of commands. A survey of spacecraft operators and instrument scientists quickly identified a short list of events that can be used to trigger most mission operations. Most events for many Earth-orbiting spacecraft are based upon geographical location, for example, *turn on the transmitter when the spacecraft is over the ground station* or *change the filter wheel position when the spacecraft is over the polar region.* Other events are based upon the relative positions of the spacecraft, the Earth, and other celestial bodies, for example, *change imager sensitivity when the spacecraft is on the night side*, or *turn on the heaters during eclipse*, or *yaw the spacecraft 180° when the Sun begins to shine on the cold side of the spacecraft.* Some events are most easily triggered purely on the basis of time, as is customary.

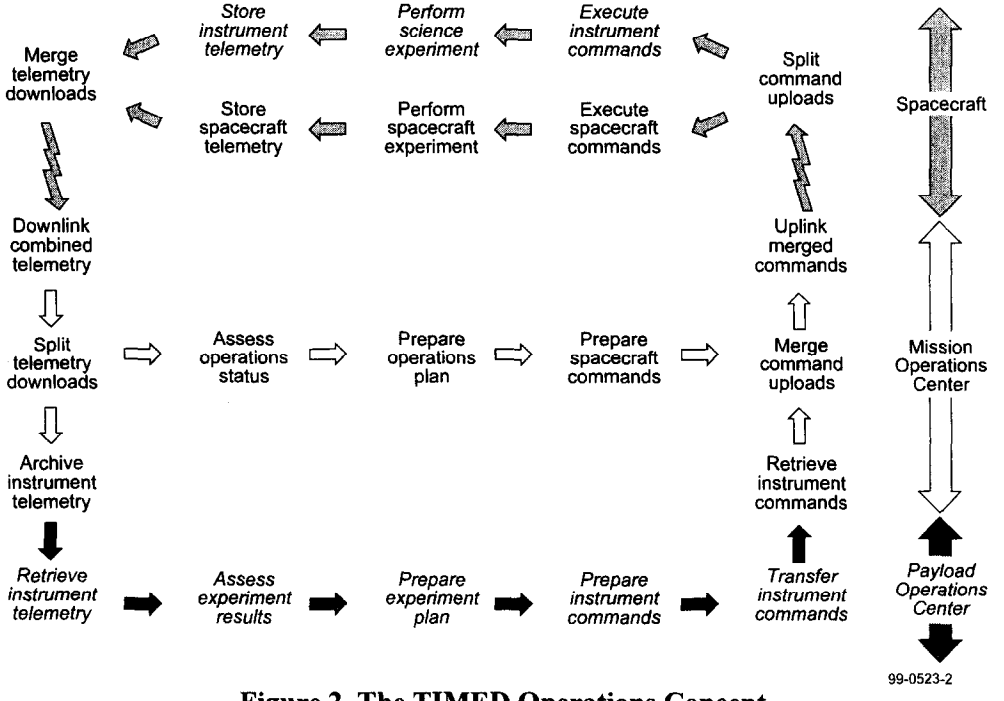


Figure 2. The TIMED Operations Concept.

Simultaneously, an effort was made to identify those ancillary data needed by scientists to interpret and correlate their data. The resulting list included parameters such as position, velocity, time, and attitude. A clear consensus emerged that many of the data needed for both event detection and ancillary data could be obtained directly using an onboard GPS receiver. While GPS is not terribly radical or new (APL first flew GPSPAC on LANDSAT-D in 1982), the GPS receiver on TIMED is being used in some very new ways.

First and foremost, GPS is being used to provide for the onboard capture of required ancillary data such as position, velocity, and time. GPS time is continuously distributed to all instruments so that science data can be accurately tagged as it is collected. All GPS data (such as position and velocity) are recorded for later playback and correlation with science data on the ground. It should be noted that, except for quality audits, these data will not be reprocessed on the ground; the data captured and recorded onboard are definitive.

These GPS data are also distributed to the instruments and spacecraft subsystems as a driver for autonomous operations. For example, based upon spacecraft position, an instrument may select a scan mode optimized to perform a polar auroral observation, or use a look-up table for geoid oblateness to limit instrument overscans for a limb scanning radiometer. GPS data can also be used to aid spacecraft operations processes. For example, GPS position can be combined with a magnetic field model to enable the Attitude Determination and Control System to use magnetic torque to dump momentum.

For TIMED, GPS is also being used to perform onboard orbit determination. Consequently, the need for performing multiple tracks at multiple ground stations using coherent transponders is completely eliminated. TIMED can save costs by performing all routine operations from a single ground station and autonomously determining its orbit on a continuously updated and refined basis. These orbit ephemerides can be transmitted to the ground and used to predict the time of future passes for automated ground station operations. Additionally, this orbit can be propagated onboard to predict ground station pass coverage on the spacecraft.

The TIMED spacecraft will routinely turn on its own downlink in a housekeeping telemetry mode whenever it is in view of its primary ground station at APL. For most passes, these data will be analyzed by an automated process to verify nominal operation and will be archived for analysis and trending. (A detection of anomalous operation can spawn a process as simple as paging the Mission Operations Manager or as complex as initiating an indexed uplink command sequence on the basis of recovered telemetry.) Once each day, however, when the ground station recovers the downlink signal and sends up the proper commands, the spacecraft will initiate its daily data recorder dump process. This process will be semi-automated initially, using a manned backup, but may eventually be fully automated to further save on operations personnel costs. It should be noted that there is a realistic lower limit which cannot be undercut or else reliable operations can no longer be assured in the event of anomalies, vacations, weekends, and sick leave. It should also be noted that

operations personnel can generally only be procured in integer-unit quantities.

The onboard orbit determination and propagation engine can be combined with GPS time to determine when the subsatellite point crosses the terminator from day to night or night to day. Several of the TIMED instruments will change measurement modes based upon the day and night flags. When GPS position and time are combined with spacecraft attitude data, a nominal Sun vector can be distributed onboard in spacecraft coordinates for instruments that want to observe the Sun or avoid it. This Sun vector may eventually be used to drive bimonthly yaw maneuvers needed to keep the Sun from shining on the spacecraft's cold side. (At 74.4° inclination and 600 km altitude, the TIMED orbit precesses three or four revolutions per year, based upon Sun-fixed or Earth-fixed reference frames.)

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